03. Exercises

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Exercise 14: A positive number

Create a program that asks the user for a number and tells if the number is positive (i.e. greater than zero).

```
Type a number: 5

The number is positive.

Type a number: -2

The number is not positive.
```

Exercise 15: Age of majority

Create a program that asks for the user's age and tells whether the user has reached the age of majority (i.e. 18 years old or older).

```
How old are you? 12

You have not reached the age of majority yet!

How old are you? 32

You have reached the age of majority!
```

Exercise 16: Even or odd?

Create a program that asks the user for a number and tells whether the number is even or odd.

```
Type a number: 2
Number 2 is even.

Type a number: 7
Number 7 is odd.
```

Hint: The number's remainder when dividing by 2 tells whether the number is even or odd. The remainder can be obtained with the `%` operator.

Exercise 17: Greater number

Create a program that asks the user for two numbers and prints the greater of those two. The program should also handle the case in which the two numbers are equal.

Example outputs:

```
Type the first number: 5
Type the second number: 3

Greater number: 5

Type the first number: 5
Type the second number: 8

Greater number: 8

Type the first number: 5
Type the second number: 5

The numbers are equal!
```

Exercise 18: Grades and points

Create a program that gives the course grade based on the following table.

Points	Grade
0–29	failed
30–34	1
35–39	2
40–44	3
45–49	4
50–60	5

Example outputs:

```
Type the points [0-60]: 37

Grade: 2

Type the points [0-60]: 51

Grade: 5
```

Exercise 19: Age check

Create a program that asks for the user's age and checks that it is reasonable (at least 0 and at most 120).

```
How old are you? 10
OK

How old are you? 55
OK

How old are you? -3
Impossible!

How old are you? 150
Impossible!
```

Exercise 20: Usernames

Create a program that recognizes the following users:

Username	Password
alex	mightyducks
emily	cat

The program should check for the username and password as follows:

```
Type your username: alex
Type your password: mightyducks
You are now logged into the system!

Type your username: emily
Type your password: cat
You are now logged into the system!

Type your username: emily
Type your password: dog
Your username or password was invalid!
```

***Exercise 21: Leap year

A year is a leap year if it is divisible by 4. But if the year is divisible by 100, it is a leap year only when it is also divisible by 400.

Create a program that checks whether the given year is a leap year.

```
Type a year: 2011
The year is not a leap year.

Type a year: 2012
The year is a leap year.

Type a year: 1800
The year is not a leap year.

Type a year: 2000
The year is a leap year.
```